Anchoring a Coordinate-Free Somatic Variant Graph: Reference-Based Localization and Geometry-Driven Reclassification of Reference-Free Structural-Variant Calls

svp project
(methodology note)

July 1, 2026

Abstract

svp calls somatic structural variants (SVs) directly from a two-colour locally-consistent-parse (LCP) core graph built from normal and tumour long reads, without assembly, alignment, or a reference genome. Its calls are therefore *coordinate-free*: each is a set of graph cores identified by hash, carrying no genomic position. We describe a final, explicitly separated pipeline stage that *anchors* these calls onto a reference coordinate system so they can be emitted as a VCF, and that uses the recovered geometry to correct systematic mis-typing produced by the coordinate-free caller. Anchoring maps each call's reference-present flanking cores to unique genomic positions, brackets the breakpoint, and then applies three signal-based operations: suppression of single-nucleotide "mis-fires" in the breakend caller, deduplication of the several redundant calls a real SV induces, and geometry-driven reclassification of large deletions and insertions that the graph caller routes through its breakend path. On a $\sim 30\times$ whole-genome simulation these operations raise precision from 4.3% to 90.7% and deletion recall from 82% to 90% at the ≥ 50 bp scale (Truvari matching), with no parameter fitted to the truth set. We further show, by extracting truth cores directly from the mutated genome, that the residual recall gap is dominated by *placement* (repeat-ambiguous flanks), not by detection: the missed variants are present in the graph and largely called, but cannot be uniquely localized.

1 Introduction

Reference-free variant calling avoids the biases of mapping reads to a single linear reference, at the cost of producing calls that live in a graph rather than on a coordinate axis. The **svp** caller builds a two-colour LCP-core graph from normal and tumour high-fidelity (HiFi) reads and emits somatic SV candidates as sets of graph cores. Downstream consumers, however, and every standard SV benchmark, expect a VCF: chromosome, position, type, and length. The gap between a coordinate-free call and a VCF record is the subject of this note.

We make three points. First, that gap is bridgeable using only the graph's own reference-present cores as anchors, with no realignment of reads (Section 3). Second, the recovered coordinates are not merely a reporting convenience: the *geometry* of a call's anchors—their spacing, and the colour of the reference cores between them—is a strong signal that the coordinate-free caller lacks, and it repairs systematic type errors that the graph alone cannot (Section 4). Third, when a reference-free caller is evaluated, the recall ceiling must be attributed carefully: we separate *detection* (is the

somatic sequence in the graph?) from *calling* (was a call emitted?) from *placement* (could the call be localized?), and find the residual loss is placement-bound (Section 6).

A methodological caveat pervades the evaluation and is reported honestly: truth "somatic cores" must be extracted from the mutated genome, not reconstructed by splicing an inserted sequence between reference flanks, or insertion recall is silently undercounted (Section 5).

2 Background: the coordinate-free caller

We summarize only what the anchoring stage consumes; the graph construction is prior work in the pipeline.

Two-colour LCP-core graph. A locally-consistent parse partitions a sequence into variable-length *cores* whose boundaries depend only on a local window, so the same locus yields the same core in any context. svp runs at level $\ell = 5$ (median core ≈ 150 bp; adjacent level-5 cores overlap ≈ 85 bp). A node's identity is $\mathrm{sig}(c) = \mathrm{XXH3\text{-}64}(c)$, a 64-bit hash of the core's bytes. Reads from both samples are parsed on both strands, counted through a per-colour Bloom cascade with a multiplicity floor, and survivors become nodes. Each node v carries a colour

$$\chi(v) = \begin{cases} 3 & \text{present in normal and tumour (shared),} \\ 1 & \text{normal-only,} \\ 2 & \text{tumour-only (candidate somatic).} \end{cases} \tag{1}$$

and per-base normal/tumour coverage. Edges join adjacent cores. The graph is coordinate-free: node payload is core *sequence*, never genome position.

Somatic calls. A somatic allele must involve a colour-2 core. The caller emits three kinds of call, each a node set:

- **BUBBLE** — a two-arm bubble with a tumour arm; typed SNP/INS/DEL from the signed spliced arm-length difference.
- **BND** — a tumour-only run bridging two shared anchors at near-full depth (a novel adjacency); intended for DUP/INV/large-indel/translocation.
- **TIP** — a tumour-only dead-end chain (insertion-end candidate).

By design the caller is high-recall: it emits *every* structure carrying a tumour-only allele and defers false-positive control downstream. On our benchmark it emits 294,740 calls (262,951 bubbles, 19,676 breakends, 12,113 tips) from a graph of $K = 86{,}519{,}562$ nodes. Each call also records the XXH3-64 signatures of its alt-allele cores.

3 Reference anchoring

Anchoring is an explicit, opt-in stage, `svp anchor <dir> <ref.fasta>`. It is deliberately *not* part of the reference-free `run` pipeline: detection stays reference-free, and a reference is consulted only to name coordinates for reporting. The reference FASTA is the sole external input; the mutated/tumour assembly is never used as a caller input.

3.1 Anchor cores

A somatic call is built from colour-2 (tumour-only) cores, which are by definition absent from the reference and therefore unlocalizable on their own. Their graph neighbours, however, of colour 1 or

3 are present in the normal genome and hence in the reference. For a call with alt-node set A we take as its *anchor set*

$$\mathcal{N}(A) \;=\; \{\, w \in \mathcal{G} : \chi(w) \in \{1,3\},\; w \notin A,\; \exists\, a \in A \text{ with } (a,w) \text{ or } (w,a) \in E \,\}. \tag{2}$$

3.2 Reference core-position index

We stream the reference once, parsing each contig on both strands at level ℓ, and record, for every needed anchor signature, its forward-genome position. Reverse-strand cores are mapped back to forward coordinates. A signature seen at more than one distinct position is *repeat-ambiguous* and dropped, yielding a partial map

$$\pi :\ \text{sig} \;\rightharpoonup\; (\text{contig}, \text{position}), \tag{3}$$

that is defined only on cores with a unique genomic location. On the benchmark, 582,109 distinct anchor signatures are required; 527,048 receive a unique position and 36,107 are dropped as repeat-ambiguous.

3.3 Breakpoint bracketing

For each call we look up π on its anchor set, keep the consensus contig (the one carrying the most located anchors), and bracket the variant between the leftmost and rightmost anchor. A subtle but important correction: an anchor's *start* coordinate lies $\approx$ one core width ($\sim$150 bp) upstream of the true breakpoint, because the breakpoint sits just past the left flank core. We therefore place the left coordinate at the left anchor's *end* (start + core length), removing a systematic -170 bp offset (measured median) that would otherwise defeat overlap-based matching for small SVs. A call all of whose anchors are repeat-ambiguous cannot be placed; 20,186 calls (7.4%) are left unanchored.

4 Signal-based refinement of anchored calls

Coordinates unlock three operations the coordinate-free caller cannot perform. Let a breakend call have anchor span s (reference distance between its flanking anchors) and tumour junction length j (the spliced length of its tumour-only run).

4.1 Suppression of single-nucleotide mis-fires (LocalSub)

A single-nucleotide variant perturbs the graph locally; its tumour-only cores can bridge two *consecutive* reference anchors, which the breakend path then reports as a novel adjacency. Such a "breakend" does not skip or rearrange the reference—its junction *reproduces* the spanned reference:

$$\text{mis-fire} \;\iff\; j \;\geq\; s/2. \tag{4}$$

The two regimes separate cleanly (Table 1): true DUP/INV breakends have a short junction relative to their span (median $j/s = 0.02$), whereas mis-fires reproduce the span (median $j/s = 1.04$). Flagged calls are marked `FILTER=LocalSub`; on the benchmark this removes 7,666 spurious breakends.

	median j/s	$j/s \geq 0.7$ (local)	$j/s < 0.3$ (real skip)
true DUP/INV breakends	0.02	6%	84%
false breakends	1.04	95%	4%

Table 1: Junction-to-span ratio separates single-nucleotide mis-fires from true novel adjacencies.

4.2 Locus deduplication

A single real SV induces several calls: a bubble *and* a breakend, each doubled by the forward/reverse-complement parse. Under one-to-one benchmark matching, one becomes a true positive and the rest false positives. We collapse calls of the same type whose anchored positions fall within a locus window, keeping the highest-support representative. This removes 43,767 redundant records (15.9%) and, as shown below, is the single largest precision gain.

4.3 Geometry-driven reclassification

Large deletions and insertions do not form clean bubbles (their reference arm branches through repeats), so the caller routes them through the breakend path and mis-types them. The anchor geometry recovers the correct type.

Deletion. A somatic deletion removes reference sequence present in the normal genome. Hence the reference cores spanning $[\,\mathrm{lo}, \mathrm{hi}\,]$ between the two anchors are predominantly *normal-only* (colour 1) in the graph, whereas a duplication's span is shared (colour 3) and an artefact's is mixed. This test is strand-agnostic. For a two-anchor breakend with $50 \leq s \leq 50{,}000$ and $j < s/2$ we fetch the spanned reference, parse it, and retype to **DEL** of size s when

$$\frac{\#\{\text{spanned cores with } \chi = 1\}}{\#\{\text{spanned cores in graph}\}} \;\geq\; 0.6. \tag{5}$$

Insertion. A somatic insertion places novel sequence between two *adjacent* reference positions: its two flanking anchors are essentially coincident ($s < 50$) while its junction is long ($j \geq 50$). We retype such calls to **INS** of size $\approx j - s$ *before* the LocalSub test, which would otherwise absorb them (a long junction trivially satisfies $j \geq s/2$).

5 Evaluation protocol

Data. A human reference is mutated by insilicoSV to a truth genome and truth VCF (125,830 variants); HiFi reads are simulated by Badread (normal from the reference, tumour from the truth genome) at $\sim 30\times$ whole-genome depth; svp is run at multiplicity floor 3. The ≥ 50 bp benchmark set contains 485 SVs (240 DEL, 78 INS, 167 DUP/INV).

Matching. We benchmark with Truvari (sequence comparison disabled, as our symbolic alleles carry no inserted sequence; breakend distance 1000 bp; `PASS`-only). Because the caller does not sub-type DUP versus INV, we relabel the truth DUP/INV records to the breakend type so that matching is type-consistent; deletion and insertion types are unchanged and *not* matched across type. No leniency (no type-ignoring) is used.

configuration	recall	precision	F_1
anchored calls (no refinement)	77.1%	4.3%	0.08
+ LocalSub mis-fire filter	75.5%	37.9%	0.51
+ locus deduplication	74.2%	84.3%	0.79
+ BND→DEL retyping	78.4%	90.9%	0.84
+ BND→INS retyping (final)	78.4%	90.7%	0.84

Table 2: Contribution of each anchoring operation at the ≥ 50 bp scale (Truvari; 485 truth SVs). No parameter is fitted to the truth set.

class	recalled / truth	recall
DEL	216/240	90.0%
INS	56/78	71.8%
DUP/INV	108/167	64.7%
all ≥ 50 bp	380/485	78.4%

Table 3: Final per-class recall (precision 90.7% overall, 380 TP / 39 FP).

Detection accounting. To attribute the recall ceiling we build, per truth variant, the set of its *out-only* cores (present in the tumour genome, absent from the reference) and test membership in (i) the read graph and (ii) the emitted call set. We stress a construction pitfall: reconstructing an insertion's cores as [reference flank]+[inserted sequence]+[reference flank] yields junction cores that do not match the tumour haplotype when other variants perturb the flanks, so the insertion appears to produce *zero* somatic cores. Extracting the cores from the mutated genome directly removes the artefact: all sampled "zero-core" insertions in fact produce 6–10 novel cores, all present in the read graph.

6 Results

6.1 Ablation

Table 2 adds each operation in turn. Anchoring alone gives high recall but catastrophic precision, because every real SV is over-called and the breakend path mis-fires on the dense SNPs. LocalSub and deduplication together lift precision from 4.3% to 84.3%; geometry-driven retyping recovers deletions (and their precision) and slightly raises insertion handling, reaching 90.7% precision at 78.4% recall.

6.2 Per-class performance

Table 3 gives the final per-class recall. Deletion recall reaches 90%; DUP/INV and insertion remain lower for reasons attributed below.

6.3 Detection versus placement

The residual recall gap is not a detection failure. Of the 24 missed deletions, 18 are *called* (an out-only core appears in an emitted call) but cannot be placed because their flanking anchors are repeat-ambiguous; only 6 are true calling misses. Of the 22 missed insertions, 21 likewise have no

anchored call near their locus despite being detected in the graph. Deletion cores are detected at 97.2% (280/288) and insertion cores, measured against tumour-genome-derived truth, at essentially 100%. The ceiling is therefore *placement*: variants whose flanks fall in repeats have no unique coordinate for any reference-based anchoring to assign.

7 Discussion and limitations

Coordinates repair type, not just report it. The most transferable observation is that reference geometry is an independent signal channel. The same anchor spacing that yields a VCF position also distinguishes a deletion (normal-only span) from a duplication (shared span) and a point insertion (coincident anchors, long junction) from a substitution (reproduced span). A coordinate-free caller cannot access these, so a modest, principled anchoring stage recovers accuracy that no amount of graph-only reasoning would.

The placement wall. Because anchoring relies on uniquely-locatable flanking cores, variants embedded in repeats are lost even when fully detected and called. Raising deletion and insertion recall past $\sim 90\%$ would require multi-position anchoring—retaining a repeat-ambiguous anchor's several candidate positions and placing a call where its anchors mutually agree—at some risk to precision. This is the natural next step and is left as future work.

Sub-typing. DUP versus INV is not resolved; both are reported as breakends. Distinguishing them requires the oriented order of the two anchors (forward for deletion, reversed for tandem duplication, opposite strand for inversion), which in turn requires tracking anchor strand through the dual forward/reverse-complement parse.

Honest accounting. Aggregate recall on this data is dominated by SNPs (94% of truth variants) and is not informative about SVs; per-class, size-stratified figures are the only meaningful ones. Truth cores for insertions must be taken from the mutated genome, or insertion detection is underreported—an artefact we encountered and corrected.

8 Conclusion

We localize the coordinate-free somatic calls of a reference-free SV caller onto a reference using only their graph-present flanking cores, and show that the recovered geometry both produces a VCF and repairs the caller's systematic mis-typing. On a whole-genome simulation the anchoring stage lifts ≥ 50 bp precision from 4.3% to 90.7% and deletion recall to 90% without any truth-fitted parameter, and it localizes the remaining ceiling to variants in repeats that no reference-based method can uniquely place. Detection is not the bottleneck; placement is.